

ConvertPro - Project Architecture Report

Overview

ConvertPro is a modern, full-stack image processing web application built with a React (Vite) frontend and a Node.js (Express) backend. It features dynamic dark mode, glassmorphism UI design, and 12+ professional image processing tools.

Frontend Architecture (React / Vite)

The frontend uses TailwindCSS for styling and Axios for API communication.

Core Files

- **frontend/src/main.jsx**: The entry point of the React application.
- **frontend/src/App.jsx**: Configures React Router, mapping `/` to the Home page and `/tool/:id` to the dynamic ToolPage.
- **frontend/src/index.css**: Contains global Tailwind directives and custom CSS variables for the light/dark themes and animations.
- **frontend/tailwind.config.js**: Configures the Tailwind theme, extending colors (using `#4A90E2` for the primary accent) and enabling `class` based dark mode.

Pages

- **frontend/src/pages/Home.jsx**: The landing page that displays a responsive grid of `ToolCard` components for all available image tools.
- **frontend/src/pages/ToolPage.jsx**: The core dynamic engine. It reads the URL parameter (e.g., `crop`, `ocr`) and dynamically renders the appropriate UI settings (sliders, text inputs) and handles API routing based on the selected tool.

Components

- **frontend/src/components/ui/UploadArea.jsx**: A reusable Drag & Drop component that handles file selection and dynamically injects custom tool configurations before submission.
- **frontend/src/components/ui/ToolCard.jsx**: A stylistic card component used on the Home page.
- **frontend/src/components/layout/Navbar.jsx**: Contains the logo and the global Dark/Light mode toggle switch.
- **frontend/src/components/layout/Footer.jsx**: Displays copyright and credits (Website by HEMU).

Backend Architecture (Node.js / Express)

The backend leverages `multer` for multipart form data uploads and powerful processing libraries like `sharp`, `tesseract.js`, and `potrace`.

Core Files

- **backend/server.js**: The main Express server file. It configures CORS, initializes the server on port 5000, and mounts all the individual router modules.

API Routes

- **backend/routes/convert.js**: Handles format conversions (JPG, PNG, WEBP). Supports batch uploads and uses `archiver` to return a ZIP file.
- **backend/routes/compress.js**: Uses `sharp` to reduce image quality and file size based on a user-defined percentage.
- **backend/routes/edit.js**: Contains endpoints for Resizing (width/height), Cropping (x/y coordinates), and Rotating/Flipping images natively.
- **backend/routes/pdf.js**: Stitches multiple uploaded images into a single PDF document.
- **backend/routes/advanced.js**: The powerhouse module containing 6 advanced features:
 - `/exif`: Extracts hidden EXIF metadata via `exifr`.
 - `/strip-exif`: Uses `sharp` to remove all metadata.
 - `/upscale`: Uses `sharp`'s Lanczos3 kernel to double or quadruple image resolution.
 - `/to-svg`: Traces pixels into scalable vectors using `potrace`.
 - `/favicon`: Generates a ZIP of web-ready `.ico` and `.png` icons in standard sizes.
 - `/ocr`: Extracts English text from screenshots using `tesseract.js`.
 - `/make-gif`: Compiles multiple images into an animated GIF using `gif-encoder-2`.